A VISIT TO CERN · 23 JUNE

My CERN Diary

Tunnels, magnets, quantum puzzles and one very big question, written down so I never forget them.

Discover CERN · Our Universe · Quantum World · Exploring the Unknown

by Vidika Mirpuri



Outside the ATLAS building, with the detector painted on the wall behind me

Numbers I want to remember

27 km — length of the LHC ring

1.9 K — magnet temperature, colder than space

92 million — pixels in ATLAS's innermost detector

1957 — the Synchrocyclotron switches on

5% — of the universe is ordinary matter

Inside this diary

- Meeting ATLAS — ATLAS Visitor Centre
- CERN's very first accelerator — The Synchrocyclotron
- Walking up to the LHC — Discover CERN
- The biggest questions — Our Universe
- A glowing silhouette — Quantum World
- Inside a detector — Exploring the Unknown

ENTRY 1 ATLAS VISITOR CENTRE

Meeting ATLAS, the detector that catches collisions

Our visit began at the ATLAS Visitor Centre. ATLAS is one of the giant detectors on the Large Hadron Collider, built to catch what happens when protons smash into each other. Our guide stood beside a huge screen and took us through it layer by layer, like unpeeling an onion made of metal and silicon.

The innermost layer is the pixel detector, only 3.3 to 12 centimetres from the beam line. It has four layers of silicon pixels, and each pixel is smaller than a grain of sand. Further out, the calorimeters stop the particles and measure how much energy they carry. The liquid-argon one is chilled to about −185 °C.

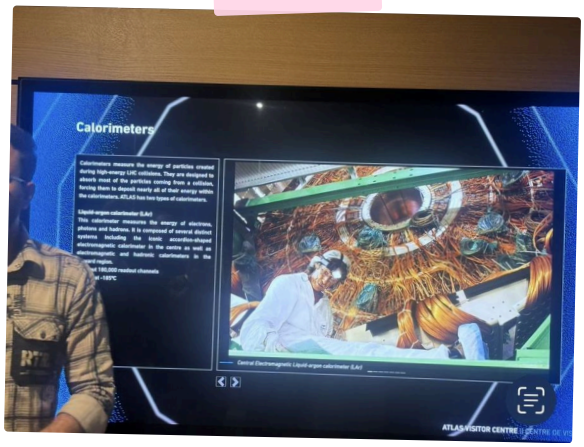
Through a window we could see the ATLAS Control Room, with rows of screens and people keeping watch over the detector. It looked like a very serious computer lab, and I loved that something so huge is looked after from a room like that.

What I learnt

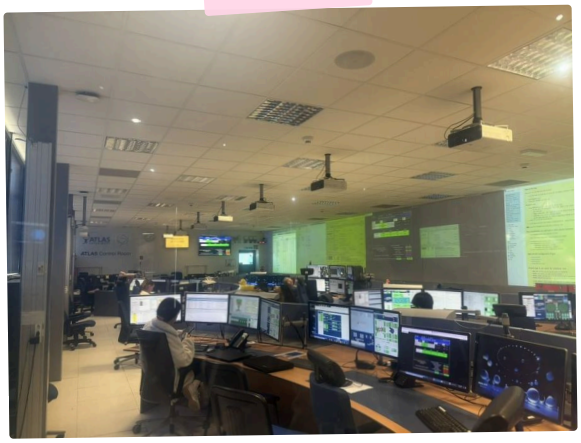
- The inner detector measures the direction, momentum and charge of electrically charged particles.
- The pixel detector has 92 million readout channels, a bit like a 92-megapixel camera.
- Calorimeters make particles deposit nearly all their energy so it can be measured.
- Detectors are built in layers because each layer answers a different question.



The Inner Detector, explained on screen



Calorimeters: where particles hand over their energy



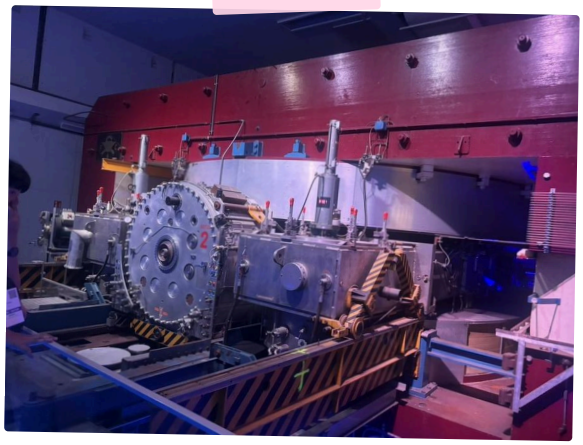
The ATLAS Control Room, seen through the glass

ENTRY 2 THE SYNCHROCYCLOTRON

Visiting CERN's very first accelerator

Next we stepped into a blue-lit hall with an enormous red magnet and a tangle of metal drums, pipes and warning stripes. This is the Synchrocyclotron, CERN's first accelerator, which started running in 1957. Standing behind the glass, I tried to imagine physicists working around it back then.

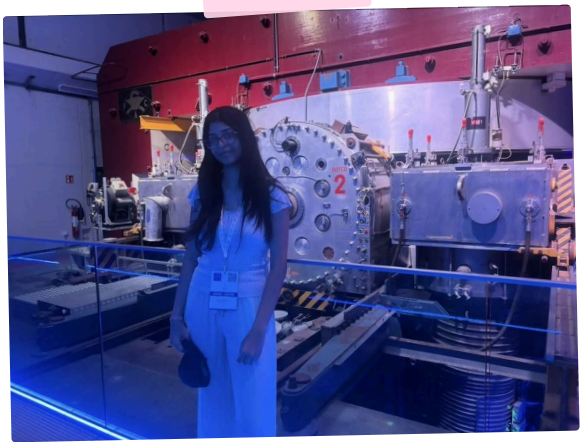
In a corner, black-and-white portraits of scientists looked down from the wall above round alcoves, and a chalkboard still carried physics written out by hand. It felt like the room had been paused in the middle of an experiment.



The Synchrocyclotron hall

What I learnt

- The Synchrocyclotron was CERN's first accelerator, switched on in 1957.
- In a machine like this, a huge magnet bends the particles around while electric fields give them more energy.
- From this hall-sized machine, accelerators grew into the 27 km ring of the LHC.



Me, in front of the machine where CERN's story began



The blue corner with portraits and a chalkboard

ENTRY 3 DISCOVER CERN

Walking up to the Large Hadron Collider

This exhibition made the LHC feel real. There is a full-size piece of the machine you can walk right up to, painted bright blue, sitting in a tunnel-like space with pipes and cables overhead. Nearby, a cutaway let me peek inside through small windows.

A touchscreen showed a slice through one of the magnets. I tapped through the labels one by one: cryostat, helium pipe, collars, yoke, magnetic coils, support posts, and in the middle the vacuum chamber where the beams travel. I kept going between the real magnet and the screen, matching each name to a part.



A full-size section of the LHC, in bright blue



Tapping through the parts of a magnet



A cutaway of the machine, with a screen beside it

What I learnt

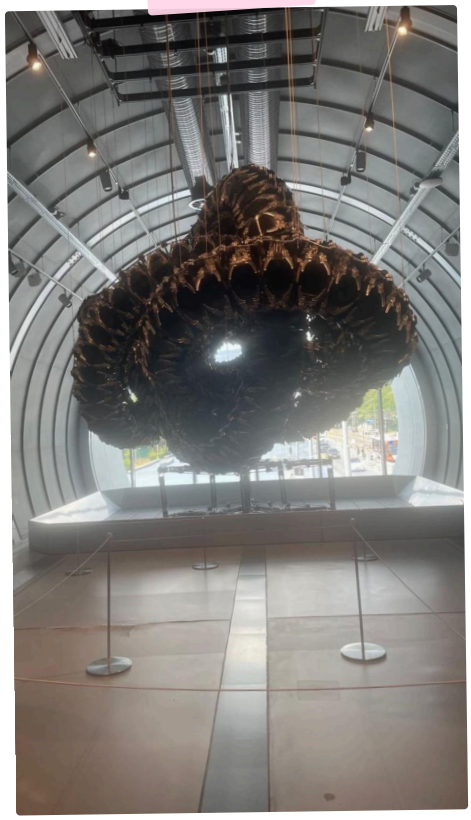
- The LHC is a 27 km ring buried underground, crossing the French–Swiss border.
- Its magnets are cooled to about 1.9 K (−271 °C) with superfluid helium, colder than outer space.
- Two beams of protons travel in opposite directions inside an ultra-high vacuum.
- Magnetic coils steer the beams, and the beams are made to collide inside detectors like ATLAS.

ENTRY 4 OUR UNIVERSE

A dark sculpture and the biggest questions

Under a tall arched roof hung a huge dark sculpture, a cloud of curved black shapes with a hole in the middle where daylight shone through. I could not stop looking at it. Our Universe is about the biggest questions there are: where everything comes from and what it is all made of.

The thing that stayed with me is how little we can actually see. All the stars, planets and people are only a small slice of the universe, and the rest is something we have not yet understood.



The hanging sculpture, with daylight through its centre

What I learnt

- Ordinary matter is only about 5% of the universe. About 27% is dark matter and about 68% is dark energy.
- The LHC recreates the conditions of the tiniest moments after the Big Bang, about 13.8 billion years ago.
- The Big Bang should have made equal amounts of matter and antimatter, so why is our universe made of matter? Nobody knows yet.

ENTRY 5 QUANTUM WORLD

Turning into a glowing silhouette

This one felt like stepping inside a music video. In a curved room, with headphones on, our silhouettes were traced in neon pink, green and white across the walls, and sparkles scattered over the floor wherever someone moved.

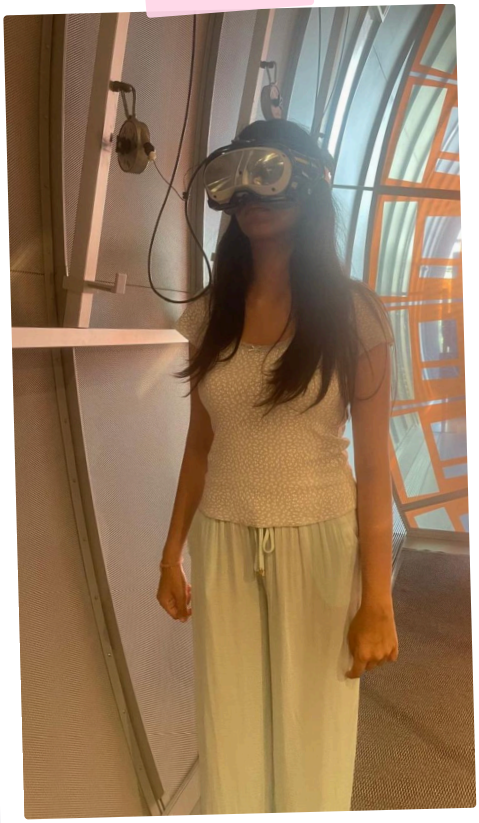
I also tried a headset with goggles and headphones. Everything about this exhibition felt a little unreal, which suits quantum physics, because the rules for very small things really are unlike anything in everyday life.



Headphones on, silhouettes glowing all around us

What I learnt

- Tiny things can act like particles and like waves at the same time (wave–particle duality).
- A quantum object can be in a mix of states until it is measured (superposition).
- Two particles can be linked so that measuring one tells you about the other (entanglement).
- Lasers and computer chips both depend on quantum physics.



Trying the goggles and headphones

ENTRY 6 EXPLORING THE UNKNOWN

Inside a detector, and a tube puzzle

Exploring the Unknown felt like walking inside a particle detector. A tunnel of glass and steel rings led to a glowing disc at the far end, and I stopped in it for photographs more than once.

Further in, a bright yellow stand held a detector module, a tiny microscope in a glass box and a tablet showing what particle tracks look like. Opposite it was my favourite station, a rack of clear tubes labelled A to F. I placed balls in at the top and watched which ends glowed blue: D, E and D on the three shelves.

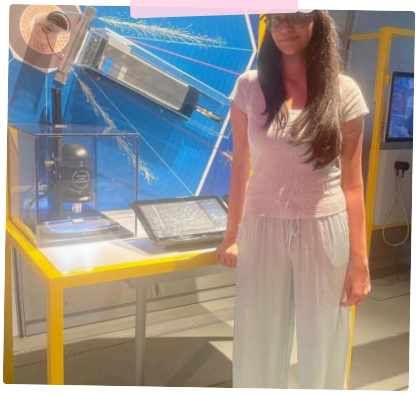
The big idea of the whole room is that detectors tell us about particles we can never see directly. We read their tracks, like detectives.

What I learnt

- A magnetic field bends the tracks of charged particles, and how much they bend reveals their charge and momentum.
- Layers of detectors record a particle's path and its energy.
- Scientists search for what they do not yet understand, such as dark matter and antimatter.



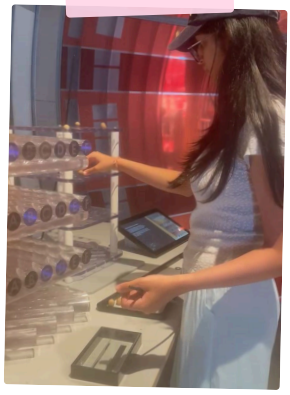
The detector tunnel



A detector piece, a microscope and a tablet



The A to F tube station



Playing with the tubes



Another look down the tunnel

What I am taking home

I arrived knowing CERN as a name from textbooks. I left understanding that it is people, machines and curiosity stacked in layers, from a 1957 magnet in a blue-lit hall to detectors that record collisions tens of millions of times a second.

The lesson I want to keep is that the most exciting part of science is the part we have not figured out yet. Dark matter, dark energy and the missing antimatter are still open. Maybe one day I will help answer one of them.

Dear future me, keep asking questions.